

Department of Information Technology (Bsc.IT Semester III)
Python Programming Practical

Practical – 10

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Practical No. 10 - Write Algorithm, Code and Output

Practical Assignment -10

Q. 1. Write algorithm for the following:

a. Write a program that compares two dates (in DD/MM/YYYY format) and prints which one is earlier.

```
from datetime import datetime

# Input from user
date1 = input("Enter first date (DD/MM/YYYY): ")
date2 = input("Enter second date (DD/MM/YYYY): ")

# Convert string to datetime object
d1 = datetime.strptime(date1, "%d/%m/%Y")
d2 = datetime.strptime(date2, "%d/%m/%Y")

# Compare and print result
if d1 < d2:
    print(f"{date1} is earlier than {date2}")
elif d1 > d2:
    print(f"{date2} is earlier than {date1}")
else:
    print("Both dates are the same.")
```

```
Enter first date (DD/MM/YYYY): 1/9/2026
Enter second date (DD/MM/YYYY): 2/3/2026
2/3/2026 is earlier than 1/9/2026
|
```

b. Write a program to measure program execution time.

```
import time

# Record start time
start_time = time.time()

# sum of numbers upto 1000000
total = 0
for i in range(1, 1000000):
    total += i
print("Sum of numbers upto 10000: ", total)

# Record end time
end_time = time.time()

# Calculate execution time
execution_time = end_time - start_time
print("Execution Time:", execution_time, "seconds")

Sum of numbers upto 10000: 499999500000
Execution Time: 0.17898845672607422 seconds
|
```

c. Write a program using the calendar module to print the weekday of the first day of a given month and year.

```
import calendar

# Input from user
year = int(input("Enter year (YYYY): "))
month = int(input("Enter month (1-12): "))

# Get weekday of the first day of the given month and year (0=Monday, 6=Sunday)
weekday = calendar.weekday(year, month, 1)

# List of Weekday names
days = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"]

# Print Result using formatted string
print(f"The first day of {month}/{year} is a {days[weekday]}")
```

```
===== RESTART: C:/Users/S:
Enter year (YYYY): 2026
Enter month (1-12): 11
The first day of 11/2026 is a Sunday

===== RESTART: C:/Users/S:
Enter year (YYYY): 2024
Enter month (1-12): 1
The first day of 1/2024 is a Monday
|
```

Q. 2. Write a python code snippet to calculate the number of days between two dates.

```
from datetime import datetime

date1 = input("Enter first date (DD/MM/YYYY): ")
date2 = input("Enter second date (DD/MM/YYYY): ")

d1 = datetime.strptime(date1, "%d/%m/%Y")
d2 = datetime.strptime(date2, "%d/%m/%Y")

days_difference = abs((d2 - d1).days)

print(f"Number of days between {date1} and {date2}: {days_difference}")

===== RESTART: C:/Users/Shubham Gawde/
Enter first date (DD/MM/YYYY): 1/9/2026
Enter second date (DD/MM/YYYY): 9/11/2026
Number of days between 1/9/2026 and 9/11/2026: 69
```

Q. 3. Explain date and time with an example?

Q. 4. How to stop program execution temporarily in python using sleep()?

